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The Role of Speed, Change of Direction, and Momentum by Position and Starting Status in Division 1 Collegiate Football Players

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Abstract

Mann, JB, Cowley, N, and Weakley, J. The role of speed, change of direction, and momentum by position and starting status in Division 1 collegiate football players. *J Strength Cond Res* 39(1): 41–47, 2025—This study (a) investigated differences between big, mid, and skill positions in sprint and change of direction times and momentum; (b) compared starting and nonstarting athletes; and (c) investigated whether thresholds can be developed to distinguish between starting and nonstarting Division 1 collegiate football athletes. Data from 496 collegiate football players who completed the 40-yard dash, pro-agility, and L drill were analyzed. Momentum was calculated using body mass and the average velocity during each test. To assess differences between positions and starters and nonstarters, data were analyzed using linear mixed models with effect size ±95% confidence intervals. Receiver operating characteristic (ROC) curves were generated to determine whether a cutoff value could be used to distinguish starters from nonstarters. Significant differences for both time and momentum were found between positional groups and starters and nonstarters for all tests in all positions. Starting skill position players tended to have greater differences in sprint or change of direction times and starting big players had greater sprint momentum. However, it should be noted that all ROC curves demonstrated relatively poor predictive value. Collectively, these findings demonstrate that bigger, faster players are preferentially selected in collegiate Division 1 football and there may be value in coaches collecting and assessing different outcome measures (e.g., sprint times and sprint momentum) depending on the positional group of the player. Finally, it should be acknowledged that setting binary thresholds to guide selection decisions is ill-advised and that speed, change of direction, and momentum are only one piece of the performance puzzle.

Key Words: American football, performance, team sports, sprinting, testing, strength and conditioning

Introduction

American football is an intermittent team sport, where athletes are required to sprint, rapidly change direction, tackle, and collide with opposition players (3). The sport has a range of unique positions and positional groups that have distinct physical characteristics and experience different physical demands (3). For example, "skill" positions (i.e., wide receivers, running backs, cornerbacks, and safeties) are typically lighter and faster than athletes within "big" positions (i.e., offensive linemen and defensive linemen), who are often required to block or tackle opposition players (7,14). Alternatively, "mid" positions (i.e., tight ends, quarterbacks, linebackers, and defensive ends) often require a mix between physical size and agility (7,14). With the extremely disparate demands between positional groups (11), the development of differing physical qualities is also prevalent. However, to gain a better understanding of an athlete, it is important to consider a range of physical qualities and how they may interact with each other.

To optimize physical performance, American football athletes often have well-developed physical characteristics (13). For example, athletes who compete at the highest levels of the sport are often faster and have superior change of direction (COD) ability when compared with athletes who are younger or compete at lower levels (e.g., professional vs. semi-professional athletes) (7,13). Furthermore, in collision sports such as rugby union and rugby league, the ability to physically dominate an opponent has been demonstrated to be important, with traits like sprint momentum (i.e., body mass × velocity) being an essential consideration for the long-term progression of an athlete (1,9,18). In American football athletes, sprint momentum has been shown to be a sensitive measure of monitoring performance changes across a collegiate career (12). However, the ability of momentum-based measures to discriminate between higher performing athletes (e.g., starters vs. nonstarters) is still unknown.

Previous research that has investigated the success of American football athletes within the National Football League (NFL) has demonstrated that players who go on to receive All-Pro and Pro Bowl honors often have greater mass, sprint speed, and change of direction performance (7). Furthermore, different positional playing groups (e.g., skill vs. big positional groups) may be more reliant on different physical qualities because of the match demands that they experience (19), and momentum-based outcome measures may help to discriminate between playing ability

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(e.g., starters vs. nonstarters) (1). Consequently, if these tests and outcome measures can be used to differentiate and discriminate between player position and ability, establishing thresholds that can be used by practitioners to guide subsequent training would be useful. Considering the above, this study had 3 aims: (a) To investigate the differences between big, mid, and skill positions in sprint and change of direction times and momentum in Division 1 football athletes; (b) compare starting and nonstarting athletes; and (c) investigate whether practical thresholds can be developed that can distinguish between starting and non-starting athletes for speed, change of direction, and momentum-based outcome measures.

Methods

Experimental Approach to the Problem

To compare sprint, change of direction, and momentum differences between positional groupings, compare the physical performance and characteristics of starting and nonstarting athletes and establish thresholds that can be used to guide practitioners, a retrospective analysis of 15 years of performance data from athletes in a Division 1 collegiate football program was completed. Athletes were grouped into 3 positional categories (i.e., big, mid, and skill positions) and whether they were starter or nonstarters in their first year of college. Forty-yard sprint, pro-agility shuttle, and L drill tests were used to assess sprint performance and change of direction capacity. In addition, the times from these tests were used to calculate average running velocity and multiplied by the athlete's body mass so that the momentum from each test could be calculated.

Subjects

Across a 15-year time frame, data were collected from 496 first-year National Collegiate Athletics Association Division 1 collegiate football athletes, ages 18–23 years old, who participated in the university's athletic performance training program. Table 1 presents demographics of athletes from each positional group. The athletes were evaluated after the completion of a 6-week winter strength and conditioning program that was designed to increase muscular strength, power, speed, and agility. No athletes under the age of 18 years were included in the study, and the protocol was approved by the University of Missouri Institutional Review Board in accordance with the Declaration of Helsinki. Written informed consent was obtained from all athletes.

Procedures

All tests were administered by the university's athletic performance training staff, who were Certified Strength and Conditioning Specialists. The procedures for the testing battery were consistent for each year and were conducted in a sequential manner described below (16).

Table 1
Descriptive data of included athletes.

Position	n	Body mass (kg)	Height (m)
Skill	196	85.9 ± 6.8	1.78 ± 0.23
Mid	178	98.3 ± 10.2	1.83 ± 0.15
Big	122	127.5 ± 10.9	1.88 ± 0.19

Body Mass Assessment. Each athlete's body mass was recorded using a standard physician's beam scale (Mettler Toledo, Columbus, OH). The athletes stood on the scale and were instructed to remain motionless until the beam indicator was aligned and stable with the center mark. In accordance with the university's athletic performance training program's policy, body mass was recorded to the nearest 1 lb and then converted to kilograms by dividing body mass in pounds by 2.2.

Performance Testing. All sprint and COD momentum assessments were performed on an indoor artificial turf surface (Indoor Field Turf; FieldTurf, Montreal, CA), whereas the distance for the layout of each test was measured using a steel tape measure. The athletes were instructed to dress in standard team-issued T-shirts and shorts and their cleats.

Forty-Yard Sprint and Momentum. For the 40-yard sprint assessment, the athletes completed 2 trials separated by a recovery period of at least 5 minutes. The athletes were instructed to assume a 3-point stance for the starting position (i.e., 1 hand and 2 feet in contact with the ground) (17). An electronic timing device (SpeedTrap, Model II; Brower Timing Systems, Draper, UT) was used to measure sprint times, with the athlete instructed to place one hand on the device's touchpad. Once the athlete initiated the start of their sprint, they moved their hand from the touchpad, which triggered the electronic timing device to begin timing the sprint. The timing ended once the athlete broke an infrared beam placed 75 cm above the ground at the 40-yard mark. The best effort from the 2 attempts was used in the analyses. For the calculation of average sprint momentum ($kg \cdot m^{-1} \cdot s^{-2}$), sprint time was divided by the distance in meters (m) which was used to calculate average velocity, and this was multiplied by body mass in kilograms.

Pro-agility Shuttle. During the pro-agility assessment, the athlete completed 2 trials separated by at least a 5-minute recovery period. The athletes were instructed to straddle a yard marker line on the field with one hand on the ground for the starting position. The athletes were instructed to self-select the direction of their initial movement to start the test. The athlete then ran 5 yards in the initial self-selected direction, then turned 180° as their outside hand touched a marked line. The athlete then ran 10 yards in the opposite direction until they touched the next marked line with their outside hand. Finally, the athlete ran 5 yards back through the original starting line. Two test administrators with handheld stopwatches (Model SC-5-5; Robic, Inc., Orange, CA) recorded time to completion. Athletes were required to face the administrators for each turn. The test administrators started the stopwatches based on the initial movement of the athlete and stopped timing when any part of the athlete's body crossed the finish line. The best time for 2 trials was recorded to the nearest 0.10 seconds. The average times recorded by both test administrators used as the athlete's final time. For the calculation of pro-agility momentum ($kg \cdot m^{-1} \cdot s^{-2}$), test time was divided by 18.29 m (i.e., distance covered during the test), which was used to calculate average velocity during the test, and this was multiplied by body mass in kilograms.

L Drill. Each athlete completed a minimum of 2 L drill trials separated by a minimum recovery of 5 minutes between trials. The athletes were timed by 2 test administrators with handheld stopwatches (Model SC-5-5; Robic, Inc.). The athletes were instructed to assume a 3-point stance for the starting position. The

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Table 2

Estimated marginal mean ± *SE* (95% CI) for sprint, pro-agility, L drill time, and momentum for big, mid, and skill positions.*

Positional group	40-yard sprint time (s) Estimate ± *SE* (95% CI)	Pro-agility time (s) Estimate ± *SE* (95% CI)	L drill time (s) Estimate ± *SE* (95% CI)	40-yard sprint momentum ($kg \cdot m^{-1} \cdot s^{-2}$) Estimate ± *SE* (95% CI)	Pro-agility momentum ($kg \cdot m^{-1} \cdot s^{-2}$) Estimate ± *SE* (95% CI)	L drill momentum ($kg \cdot m^{-1} \cdot s^{-2}$) Estimate ± *SE* (95% CI)
Big positions	5.24 ± 0.04 (5.15–5.33)	4.71 ± 0.04 (4.63–4.80)	7.79 ± 0.05 (7.68–7.90)	891 ± 26.6 (832–950)	495 ± 15.2 (461–529)	449 ± 14 (418–480)
Mid-positions	4.78 ± 0.03 (4.71–4.85)	4.37 ± 0.03 (4.31–4.44)	7.19 ± 0.04 (7.11–7.28)	762 ± 20.4 (716–807)	416 ± 11.68 (390–442)	380 ± 11 (356–403)
Skill positions	4.60 ± 0.03 (4.53–4.68)	4.23 ± 0.03 (4.16–4.30)	7.01 ± 0.04 (6.92–7.09)	680 ± 20.8 (633–726)	369 ± 12.0 (342–395)	334 ± 11 (310–358)

*CI = confidence interval.

trials began at the first visible movement as the athlete sprinted 5 yards straight ahead, touched a line with their right hand, performed a 180° turn, sprinted 5 yards back to the original start line, and again touched the line with their right hand. Then, the athlete immediately performed another 180° turn and sprinted 5 yards, made a 90° turn to the right and sprinted 5 yards, performed a 180° change of direction around a cone, sprinted straight ahead 5 yards, to where they turned 90° to the left, and sprinted a final 5 yards through the original start line. The best time for 2 trials was recorded to the nearest 0.10 seconds, with the average times recorded by both test administrators used as the athlete's final time. For the calculation of L drill momentum ($kg \cdot m^{-1} \cdot s^{-2}$), test time was divided by 27.43 m (i.e., the distance covered), which was used to calculate average velocity throughout the test, and this was multiplied by the subject's body mass in kilograms.

Designation of "Starter" and "Nonstarter". Starter and nonstarter designations were given to players based on the criteria that if the player started any match, they were immediately deemed a "starter." If a player did not start a single match, they were deemed a "nonstarter."

Statistical Analyses

Data are presented as estimated marginal mean ± *SE* (95% confidence intervals [95% CIs]) or standardized effect size (ES) ± 95% CI. QQ and residual plots were used to assess the normality and homogeneity of data, respectively. Linear mixed models were used to assess pairwise comparisons for time and momentum between positional groups, and starters and nonstarters. The model used for analysis incorporated positional group and career starting status as fixed effects with an interaction term between them, with individual position considered as a random effect. Thresholds for ESs were set at <0.20 (*trivial*), 0.20–0.59 (*small*), 0.60–1.19 (*moderate*), 1.20–1.99 (*large*), and ≥2.0 (*very large*) (8). If CIs crossed −0.20 and 0.20, the effect was determined to be unclear. Statistical significance was set at $p \leq 0.05$. In addition, receiver operating characteristic (ROC) curves were used to establish cutoff values that could support the identification of starters and nonstarters. These cutoff values were established through prioritizing sensitivity and positive predictive value (PPV) from the modeled ROC curve as key outcome measures, as coaches often wish to identify high-performing players. Greater sensitivity ensures that our classification model captures the majority of actual starters, whereas PPV demonstrates the accuracy of starter identification (4). By focusing on these variables, we aim to ensure the effectiveness of our classification model in meeting the primary objective of accurately distinguishing starters from nonstarters in the context of athletic performance analysis. Youden's J statistic was provided to offer an evaluation of the balance between sensitivity and specificity at the provided threshold (4). In addition, the area under the curve (AUC) value was provided as a measure of the overall discriminative power of the classification model. Model classifications for AUC were set at 0.51–0.69 (*poor*), 0.7–0.79 (*fair*), 0.8–0.89 (*good*), ≥0.9 (*excellent*) (4). Data analysis was performed within R Studio (version 4.1.2) using lme4, lmerTest, emmeans, and pROC packages (2,10,15).

Results

Descriptive positional group differences can be found in Table 2, with between positional group comparisons being found in Figure 1. Between positional groups, all differences were statistically significant, with ESs ranging from *moderate* to *very large*. Time-based and momentum-based outcomes from starters and nonstarters within each positional group can be found in Tables 3 and 4, respectively. Between starters and nonstarters within each positional group, all differences were statistically significant, with ESs ranging from *small* to *moderate*. Visual distribution of starter and nonstarter data can be found in Figure 2. Finally, ROC curve analysis thresholds and measures of the performance of the classification models in distinguishing between starters and nonstarters can be found in Table 5. The overall discriminative power of the classification models, represented by the AUC value, were *poor* to *fair*.

Discussion

The aims of this study were to (a) establish the differences between big, mid, and skill positions in sprint and change of direction times and momentum; (b) compare starting and nonstarting athletes within positional groups; and (c) investigate whether practical thresholds can be developed that can distinguish between starting and nonstarting athletes. Statistically significant *moderate* to *very large* differences in sprint and change of direction times and momentum can be observed between positional groups. However, of note are the disparities between starters and nonstarters and that the magnitude of difference varies depending on the positional group within each test. For example, in the big positions, there are greater differences in the 40-yard momentum than the 40-yard time when starters are compared with nonstarters. However, this is the opposite in the skill positions, with starters having relatively greater speed than momentum when compared with nonstarters. These differences likely occur because of the frequency of collisions and high-speed exposures players within each positional group face during match play (19). Finally, despite starting players being faster and generating more

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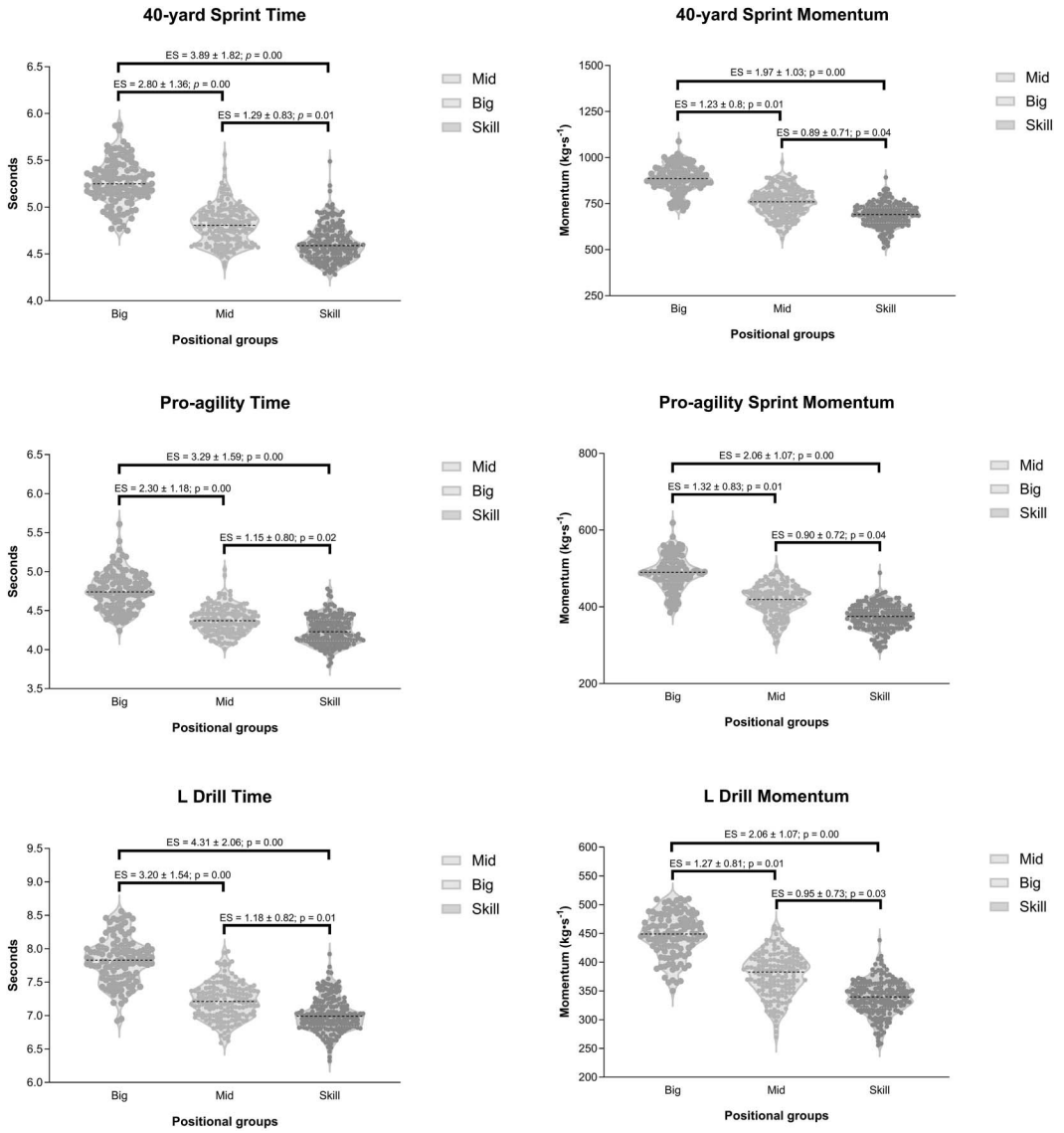

Figure 1. Violin plots with median (black dashed line), between positional group comparisons, and individual data points across 496 collegiate Division 1 American football players within big, mid, and skill positions in the 40-yard sprint, pro-agility test, and L drill.

Table 3

Estimated marginal mean ± *SE* (95% CI) for starters and nonstarters in sprint, pro-agility, L drill time, and momentum for big, mid, and skill positions.*

Positional group	40-yard sprint time Estimate ± *SE* (95% CI)	Effect size (inference); *p*	Pro-agility time Estimate ± *SE* (95% CI)	Effect size (inference); *p*	L drill time Estimate ± *SE* (95% CI)	Effect size (inference); *p*
Big positions						
Starters	5.18 ± 0.05 (5.08–5.28)	0.51 ± 0.31 (small); <0.01	4.64 ± 0.04 (4.54–4.73)	0.66 ± 0.31 (moderate); <0.01	7.70 ± 0.06 (7.57–7.82)	0.58 ± 0.33 (small); <0.01
Nonstarters	5.30 ± 0.04 (5.21–5.39)		4.79 ± 0.04 (4.70–4.87)		7.89 ± 0.05 (7.78–8.00)	
Mid-positions						
Starters	4.70 ± 0.04 (4.62–4.78)	0.66 ± 0.26 (moderate); <0.01	4.34 ± 0.03 (4.27–4.41)	0.28 ± 0.27 (small); 0.02	7.14 ± 0.05 (7.04–7.23)	0.35 ± 0.29 (small); 0.01
Nonstarters	4.87 ± 0.03 (4.79–4.94)		4.41 ± 0.03 (4.34–4.47)		7.25 ± 0.04 (7.17–7.34)	
Skill positions						
Starters	4.52 ± 0.04 (4.44–4.60)	0.69 ± 0.25 (moderate); <0.01	4.15 ± 0.03 (4.08–4.22)	0.72 ± 0.26 (moderate); <0.01	6.90 ± 0.04 (6.80–6.99)	0.66 ± 0.26 (moderate); <0.01
Nonstarters	4.69 ± 0.03 (4.61–4.76)		4.31 ± 0.03 (4.24–4.38)		7.12 ± 0.04 (7.03–7.20)	

*CI = confidence interval.

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Table 4

Estimated marginal mean ± *SE* (95% CI) for sprint, pro-agility, L drill time, and momentum for big, mid, and skill positions.*

Positional group	40-yard sprint momentum Estimate ± *SE* (95% CI)	Effect size (inference); *p*	Pro-agility momentum Estimate ± *SE* (95% CI)	Effect size (inference); *p*	L drill momentum Estimate ± *SE* (95% CI)	Effect size (inference); *p*
Big positions						
Starters	923 ± 27 (863–983)	0.64 ± 0.24 (moderate); <0.01	516 ± 16 (481–550)	0.71 ± 0.24 (moderate); <0.01	466 ± 14 (434–497)	0.38 ± 0.14 (small); <0.01
Nonstarters	858 ± 27 (799–918)		475 ± 15 (441–509)		433 ± 14 (402–464)	
Mid-positions						
Starters	789 ± 21 (743–836)	0.54 ± 0.21 (small); <0.01	427 ± 12 (400–454)	0.38 ± 0.21 (small); <0.01	390 ± 11 (366–414)	0.24 ± 0.12 (small); <0.01
Nonstarters	734 ± 21 (688–780)		405 ± 12 (379–431)		369 ± 11 (345–393)	
Skill positions						
Starters	701 ± 21 (654–747)	0.41 ± 0.18 (small); <0.01	380 ± 12 (353–407)	0.40 ± 0.19 (small); <0.01	343 ± 11 (319–368)	0.21 ± 0.11 (small); <0.01
Nonstarters	659 ± 21 (612–705)		357 ± 12 (330–384)		325 ± 11 (301–349)	

*CI = confidence interval.

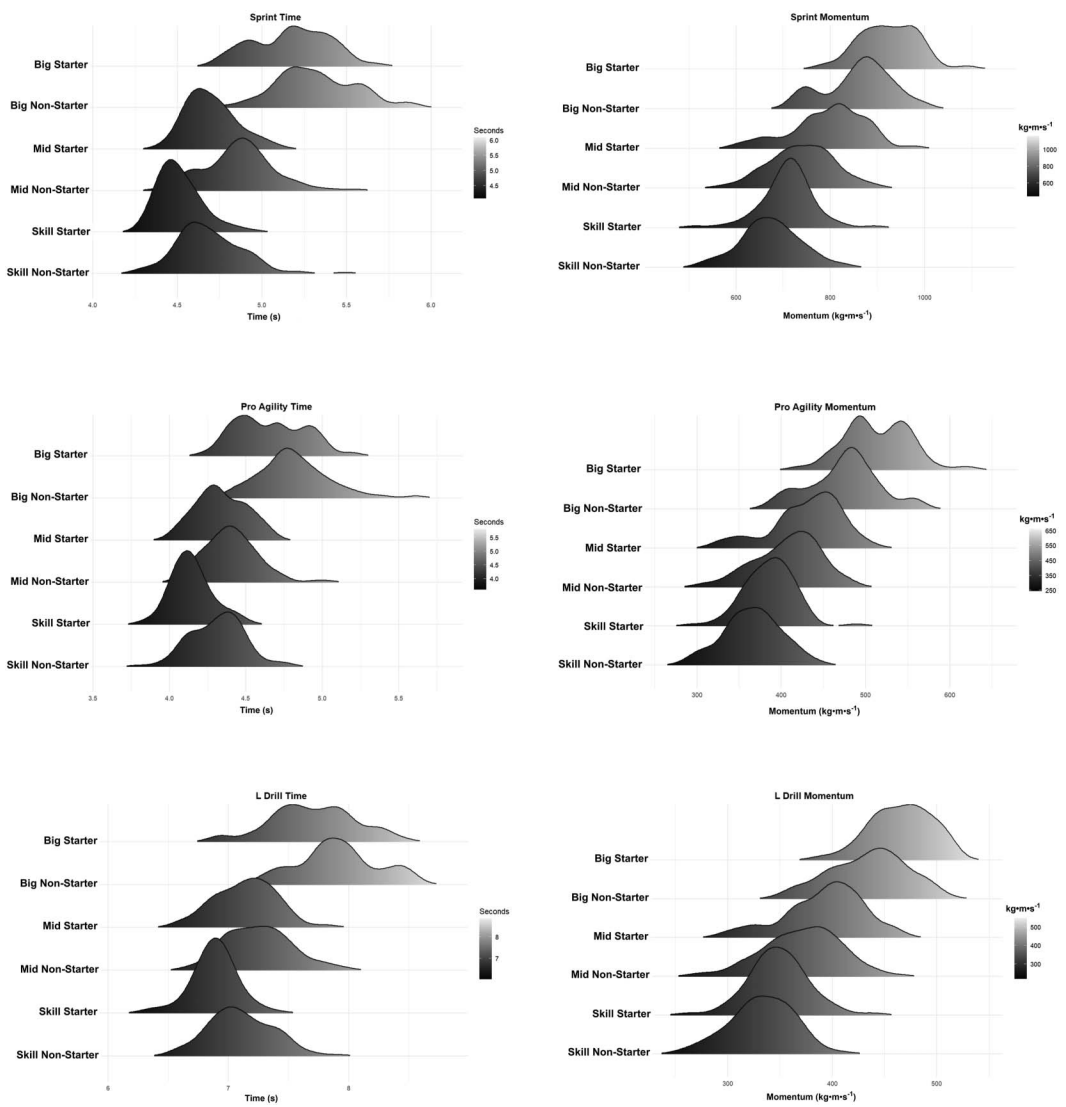

Figure 2. Ridgeline distribution of athlete data separated into positional groups and starter vs. nonstarter designation in the 40-yard sprint, pro-agility test, and L drill.

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Table 5

Receiver operating characteristic curve analysis thresholds and measures of the performance of the classification models in distinguishing between starters and nonstarters.*

	Threshold	Sensitivity (%)	PPV (%)	Youden's J	AUC; descriptor
Big positions					
40-yard sprint time (s)	5.16	61.5%	20.5%	0.33	0.66; *poor*
Pro-agility time (s)	4.72	76.9%	18.5%	0.35	0.70; *fair*
L drill time (s)	7.97	92.3%	15.0%	0.28	0.65; *poor*
40-yard sprint momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	931	77.1%	24.2%	0.39	0.74; *fair*
Pro-agility momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	492	60.4%	20.8%	0.45	0.75; *fair*
L drill momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	451	58.1%	20.0%	0.43	0.72; *fair*
Mid-positions					
40-yard sprint time (s)	4.84	75.9%	22.45%	0.25	0.65; *poor*
Pro-agility time (s)	4.36	59.3%	20.25%	0.15	0.57; *fair*
L drill time (s)	7.23	71.4%	21.98%	0.21	0.63; *poor*
40-yard sprint momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	816	84.5%	39.5%	0.36	0.73; *fair*
Pro-agility momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	454	87.9%	43.3%	0.36	0.72; *fair*
L drill momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	397	73.4%	33.9%	0.41	0.74; *fair*
Skill positions					
40-yard sprint time (s)	4.64	94.3%	27.3%	0.40	0.76; *fair*
Pro-agility time (s)	4.24	88.2%	30.6%	0.45	0.76; *fair*
L drill time (s)	7.01	87.9%	27.9%	0.41	0.72; *fair*
40-yard sprint momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	708	67.7%	30.7%	0.33	0.68; *poor*
Pro-agility momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	401	83.4%	36.6%	0.28	0.68; *poor*
L drill momentum ($kg \cdot m^{-1} \cdot s^{-2}$)	336	49.4%	24%	0.25	0.64; *poor*

*PPV = positive predictive value; AUC = area under the curve.

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momentum, thresholds that have adequate sensitivity and predictive value could not be established. Collectively, these findings suggest that coaches should use different outcome measures from the same test depending on the position of the athlete and that speed and momentum play differing roles in athlete selection depending on position. Furthermore, speed, change of direction, and momentum should be assessed in conjunction with other physical and technical or tactical abilities.

Between positional groups, there were *moderate* to *very large* differences in speed and momentum across sprinting and change of direction tests. These differences were statistically significant despite the considerable variance reported within each playing group. It is likely that the unique playing demands of each position have substantially altered the physical preparation and selection of players into certain positions (5,6,14), with bigger players who consistently produce more momentum being found in big positions, whereas smaller, faster athletes being in skill and mid positions. However, as can be seen in the variance around the ESs and what can be visually observed in Figure 1, there is considerable variation within positional groups which indicates that speed and momentum are only one aspect of the player profiling puzzle in American football.

When compared with nonstarting players, starting players were on average faster and had greater momentum when sprinting and completing change of direction tests. However, perhaps of greatest interest is the magnitude of difference between starters and nonstarters within different positional groups, tests, and outcomes. For example, there was a greater difference between skill position starters and nonstarters in absolute time compared with momentum in the 40-yard sprint, pro-agility test, and L drill. However, starters in the big positions tended to have a greater ES difference in sprint momentum outcomes when compared with assessments of time. These differences suggest that coaches should profile athletes of different positions using different outcome measures from the same test (e.g., coaches should place greater emphasis on absolute time in skill players and monitor momentum in big players). Furthermore, it demonstrates that coaches tend to pick players who are more physically developed. Previous work by Hedlund (7) noted that both greater body mass and speed are needed to make the NFL Pro Bowl or All-Pro level, and it is likely that these factors are preferentially selected at the collegiate level as well. It should be noted though that for mid-positions, the difference between sprint or change of direction times and momentum outcomes in starters and nonstarters is similar, which may be explained by the range of activities (e.g., sprinting, blocking) in their respective roles. Considering these findings, it is recommended that coaches calculate momentum from these commonly used tests and emphasize sprint and change of direction times for skill positions and consider both times and momentum outcomes for big and mid positions.

The ability to accurately predict whether a player will start or not based on their physical capacities is an appealing concept. In the current study, in a bid to provide practical thresholds, ROC curves were developed, which aimed to delineate between starting and nonstarting players for each positional group, test, and outcome measure. However, all models had *poor* to *fair* discriminative power. The thresholds selected prioritized the sensitivity (i.e., the proportion of actual starters that are correctly identified) because of the real-world need for sports coaches to identify and select promising players. But all models had relatively poor PPV (i.e., how often the model was correct when it says an athlete is a starter). This suggests that thresholds purely based on speed, change of direction, or momentum should not be used to guide selection decisions, which is likely because of the multi-faceted nature of American football that requires a range of physical qualities and technical skills (6). Thus, although starting athletes are faster and have greater momentum, it is recommended that these qualities are developed rather than used as a (de-) selection tool.

Although this is the first study to investigate the influence of speed, change of direction, and momentum characteristics by selection status across a range of positional groupings, it is not without limitations. First, despite this large dataset being

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collected over a 15-year period with nearly 500 collegiate Division 1 football athletes, it should also be acknowledged that data were attained from a single program and only a single head coach led this team over this period. Therefore, it is feasible that style of play and selection decisions may have influenced the current dataset. Second, additional ranking methods (e.g., future professional status) beyond starter vs. nonstarter were not able to be used. To support talent identification and aspirational players, identifying the qualities of players that go on to the highest level would be beneficial. Finally, to help develop tools and thresholds that may be able to guide selection decisions, moving beyond binary classification models such as ROC curves and towards methods that consider multiple variables (e.g., random forests, neural networks) may be beneficial. However, to achieve this, extremely large datasets with valid and accurate outcome measures would be essential.

In conclusion, this study demonstrates the unique sprint, change of direction, and momentum characteristics of collegiate Division 1 football players across different positional groups and starter classifications. Although players within skill positions have greater absolute sprint and change of direction times compared with mid and big positions, because of the greater mass of big positions, these players have greater momentum. In addition, across each test and outcome measure, starters were consistently faster and produced greater momentum. However, of note are the differences between time- and momentum-based outcomes for the skill positions compared with the big and mid positions. This is likely because of the greater number of collisions that big and mid position players are required to undertake compared with skill positions who complete greater high-speed distances and are regularly required to evade opposition players (e.g., wide receivers). Finally, despite the superior physical characteristics of starting players, arbitrary thresholds should not be used to select or identify possible starters. Instead, it should be acknowledged that speed, change of direction, and momentum are one important aspect of performance, and these characteristics should be developed with the support of a multidisciplinary team, alongside technical and tactical abilities.

Practical Applications

American football athletes are required to be fast, change direction efficiently, and execute these skills with substantial momentum. However, different positional groups require differing levels of these qualities. For example, skill players should focus on absolute speed and change of direction performance. Alternatively, big- and mid-position athletes need to have significant body mass to complement their running speed so that they have adequate momentum during collisions. Thus, it is recommended that the calculation of momentum during sprint and change of direction tasks is calculated to support coaches. In addition, it is acknowledged that starting players tend to have greater speed, change of direction, and momentum qualities when compared with nonstarters. Therefore, developing these qualities may be desirable and the data provided within this manuscript can be used as a reference for coaches and athletes. However, using thresholds to support selection decisions or identify potential talent is ill-advised because of the multifaceted nature of American football. Instead, the development of speed, change of direction, and momentum should be prioritized for all players, with skill positions emphasizing absolute speed and change of direction, big positions ensuring they have adequate body mass to allow for momentum to be created, and mid-position players carefully balancing the development of both speed and body mass.

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